

Digital Material Passport

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	European Research Institute Science Boulevard 42 75015 Paris FR materials@eri.example.org	Order ID PO-23456 - Pos. 1 - 2025-05-01 - Quantity 1000 kg Delivery ID DN-65432 - Pos. 1 - 2025-05-16 - Quantity 1000 kg

Product

Product Name	Titanium Alloy Ti-6Al-4V ELI	Number (UNS)	R56401
Batch ID	T-65432-01	Shape	RoundBar - D 30 mm - Length 3000 mm
Surface Condition	Machined	Product Norm	ASTM F136 (2022)
Weight	1000 kg		
Production Date	2025-05-15		
Country of Origin	DE		

Chemical Analysis

Heat Number **T-65432** - Melting Process **VAR** - Casting Method **VacuumCasting** - Casting Date **2025-05-14** - Sample Location **Product**

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
Ti	%			89.32	O	%		0.13	0.11
Al	%	5.5	6.5	6.02 ± 0.05	C	%		0.08	0.026
V	%	3.5	4.5	3.95 ± 0.03	N	%		0.05	0.012
Fe	%		0.25	0.18	H	ppm		120	35

Mechanical Properties

Property	Actual			Minimum	Maximum	Method	Status
Fatigue Test Room temperature, R = 0.1						ASTM E466	✓
Cycles (N)	10000	50000	100000	500000	1000000	10000000	
Value	650	635	610	590	570	550	
Notch Sensitivity	0.85 - 0.92			Internal Method TS-5432			✓

Physical Properties

Property	Actual	Target/Min	Maximum	Method	Status
Density	4.43 ± 0.01 g/cm³	4.43 g/cm³		ASTM B311	✓

Supplementary Tests

Test	Result / limits	Method	Status
Microstructure Examination	Equiaxed alpha with intergranular beta	ASTM E407	✓
Ultrasonic Inspection	Yes - No indications greater than reference standard	ASTM E2375	✓
Surface Quality Assessment	Class 1 - Medical Grade	Visual Inspection per ASTM F136	✓
Alpha Case Depth	5 µm - max 25 µm	Microhardness Traverse	✓
Grain Size Distribution	8 - 10 ASTM No. - target 7 - 12 ASTM No.	ASTM E112	✓

Hardness Profile (ASTM E384), values in HV

Distance from surface (mm)	0.1	0.5	1	2	3	5	10	15
Value [HV]	345	350	352	350	349	348	351	347

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.

Validated by **Dr. Markus Weber**, Head of Metallurgy, Research & Quality - 2025-05-17